GCI WORLD 2026

Transforming Churn Risk into Contract Renewals



Executive Summary

The Issue

Company A is bleeding revenue with a 49.5% baseline churn rate, heavily driven by hardware fatigue (devices > 11 months old).

The solution

A machine learning-powered "Proactive Handset Upgrade" system that targets high-risk users before they switch carriers.

Our Offer

Implementation of our 64.3% accurate XGBoost Prediction Engine, saving an estimated $1.2M annually per 10k users.

Current State of the Telecom Industry

- **The High Cost of Acquisition:** In the modern telecom sector, Customer Acquisition Cost (CAC) is dramatically higher than customer retention cost.

- **The Competitor Threat:** Competitors aggressively use subsidized hardware (free phones) as loss-leaders to break customer loyalty.

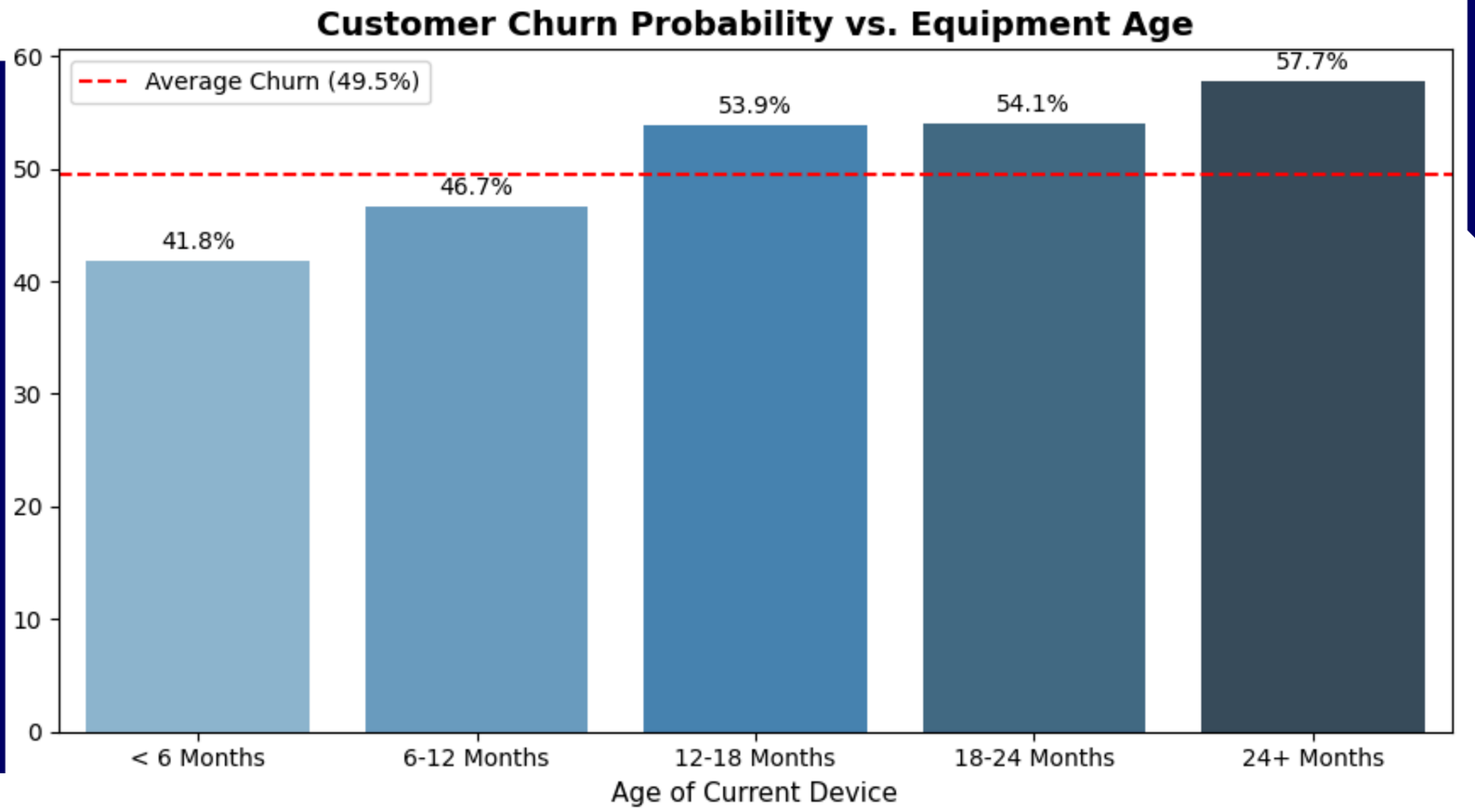
- **The Customer Mindset:** Modern customers rarely leave due to network complaints; they leave when they want a new phone and their current carrier won't subsidize it.

(Ribeiro et al., 2024; Hassouna et al., 2015)

Data Insights

Initial Exploratory Data Analysis (EDA) on network quality (dropped calls) and pricing (revenue) showed **no clear correlation to churn.**

However, segmenting by **Equipment Age (eqpdays)** revealed a massive vulnerability.



(Hunter, 2007; Waskom, 2021)

Strategic Hypotheis

The Goal

Stop customer attrition at the critical 11-month hardware cliff.

The Strategy

Reach out to customers at month 10 with a subsidized handset upgrade–conditional upon signing a new 12-month contract.

Machine Learning Methodology

Input Data

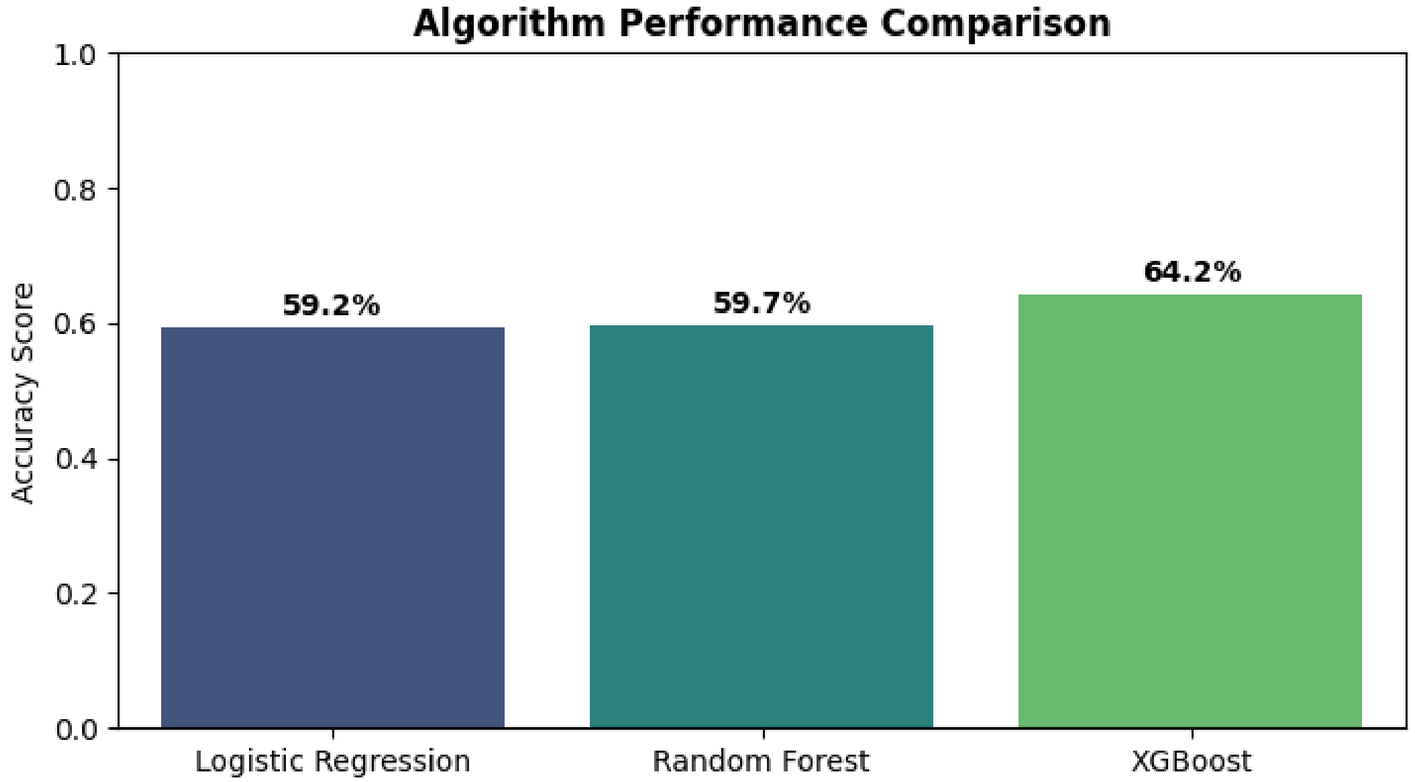
- **Dataset:** 100,000 anonymized customer records.
- **Features:** 100+ variables including demographics, usage history, billing, and equipment data.
- **Preprocessing:** Median imputation for missing values, label encoding for categories, and standard scaling.

Input Data

- **Type:** Supervised Binary Classification.
- **Target:** churn (1 = Left, 0 = Retained).
- **Objective:** Test multiple algorithms to ensure Company A receives the most accurate predictive targeting engine possible.

(Pedregosa et al., 2011)

Model Selection and Comparison



We benchmarked the industry standard (Logistic Regression) against advanced tree-based models (Random Forest and XGBoost).

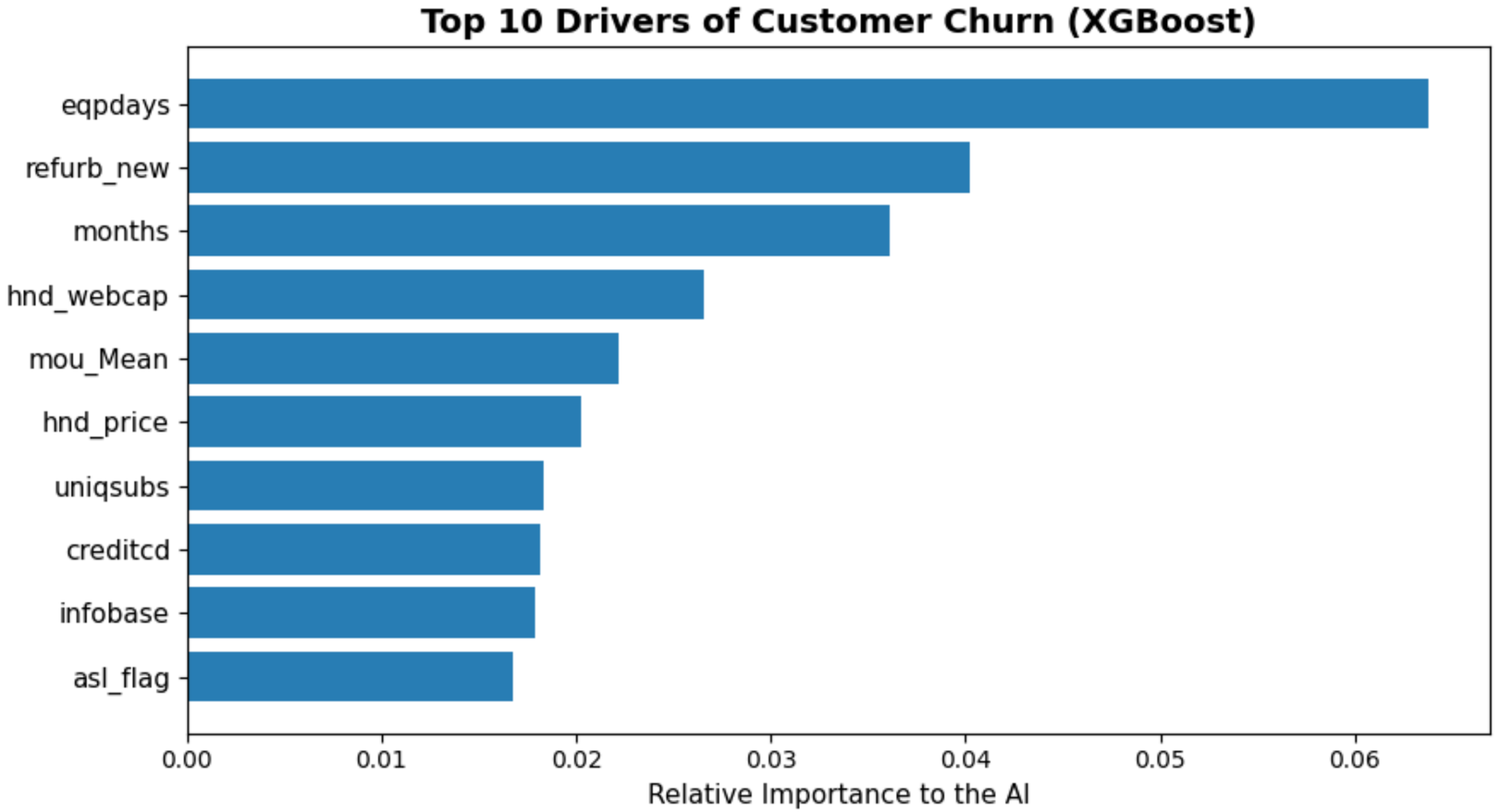
XGBoost outperformed all competitors with an accuracy of 64.2%. In a highly noisy dataset (human behavior), this represents a massive 14.8% absolute improvement over random baseline guessing.

(Hunter, 2007; Chen & Guestrin, 2016; Waskom, 2021)

Feature Validation

Our initial human-led EDA hypothesized that Equipment Age was the root cause of churn.

When we fed 100 variables blindly into the XGBoost algorithm, it independently verified our hypothesis.



(Hunter, 2007; Waskom, 2021)

Our Business Proposal

1

AI Scoring

The XGBoost model scores the entire customer base daily.

2

Target Isolation

Isolates customers with devices 300 days old who receive a "High Flight Risk" score.

3

Automated Intervention

CRM automatically triggers a targeted phone upgrade offer requiring a 12-month renewal.

Google Gemini, 2026)

Quantified Business Impact

Scenario	Outcome	Financial Impact
Cost of Inaction	~5,500 users churn (55%)	-$275,000 / month MRR lost
Intervention	Successfully retain 2,000 users	+$100,000 / month MRR retained

Return on Investment (ROI)

Assuming a $150 hardware subsidy per saved user ($300k total cost), the program pays for itself in exactly 3 months. The remaining 9 months of the contract yield pure profit.

Google Gemini, 2026)

Quotation: System Implementation

Core AI Engine Delivery

$250,000

One-Time Implementation Fee

- Custom deployment of XGBoost model to local servers.
- Full CRM and Customer Service Dashboard integration.
- Staff training on utilizing "Flight Risk Scores."

Value Proposition: This fee is a fraction of the estimated $1.2M annual revenue improvement generated from the initial 10,000-user pilot alone.

Google Gemini, 2026)

Long-Term Support & Consulting

To accelerate your business, our data science team will provide ongoing technical and strategic support to ensure the model adapts to market changes.

Monthly Retainer

$15,000 / month

Included Services

- **Model Retraining:** Adjusting the algorithm quarterly to adapt to new competitor pricing.
- **Time Series Analysis:** Linking macroeconomic indicators (inflation) to churn rates.
- **Strategic Consulting:** Monthly meetings to identify new low-risk segments for potential upselling.

Assumptions and Mitigation

1

Assumption: Hardware Costs
The ROI model assumes a $150 hardware subsidy accurately reflects Company A's wholesale vendor pricing.

2

Risk: False Positives
Because the model is 64.2% accurate, a portion of subsidies will be offered to "False Positives" (customers who would have stayed without a new phone).

3

Mitigation Strategy
The high lifetime value of retained $50/month contracts mathematically absorbs the margin lost to false positives. To optimize this, we will execute a 5,000-user A/B test in Month 1 to fine-tune the probability threshold before full deployment.



Google Gemini, 2026)

Conclusion

By combining insightful Data Analysis with advanced XGBoost targeting, we can transform Company A's biggest vulnerability (aging hardware) into a predictable, profitable contract renewal engine.

We look forward to partnering with you to secure Company A's market share.

References

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